RED SHIELD

A Quantum-Native Distributed Ledger Protocol

WHITEPAPER
Production Specifications for Public Release

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1. Abstract

Red Shield is a groundbreaking distributed ledger protocol engineered from first principles to survive and thrive in the post-quantum era. At its cryptographic heart is a **Hybrid-PQC architecture** that combines **ML-DSA-87** (NIST FIPS 204) for quantum-resistant anchors with **Ed25519** for high-performance classical signatures, ensuring the network is secure against Shor's algorithm from block zero. This security foundation is unified by the universal use of **SHAKE-256** (NIST FIPS 202) for all protocol-level hashing—including state commitments and transaction identifiers—providing 256-bit pre-image resistance and a 128-bit post-quantum security margin against Grover's algorithm.

The protocol's core innovation is **SIMBA PRIME**, a model defining five structural pillars: Self-Sovereign Node Identity (SSNI), Identity-to-Authority Binding, Merit-Based Validator Promotion, Behavioural Attestation (PoN), and Adaptive Consensus Scaling. Unlike legacy systems that rely on capital concentration or computational waste, Red Shield derives validator authority from a **Progressive Trust Score (PTS)**. This score scales through demonstrated, honest participation and is updated **atomically during block processing**, decoupling influence from financial wealth to prevent plutocratic feedback loops. Validator operational security is further hardened by the **Trinity Identity Model**, which mathematically splits node roles into a cold **Mother Wallet**, an ephemeral signing **Hot Key (SSNI)**, and a decoupled **Reward Address**.

Consensus is achieved through a Tri-Layer synergy: **Proof of Authority (PoA)** for leader scheduling, **Proof of Capacity (PoC)** for storage-based proposals, and **Proof of Network (PoN)** for multi-validator finality via an adaptive threshold that scales logarithmically with network size. This is reinforced by **Proof of Holding (PoH)** thresholds for registration and a **100-block merit gate** for status promotion. Network resilience is guaranteed by the **Never Halt** directive, a novel resolution algorithm that ensures zero transaction loss by recovering and re-including orphaned user transactions during partition recovery.

This specification bridges conceptual design with the production-ready **SIMBA_PRIME_3.1** implementation. It details machine-bound security measures including **Windows DPAPI** encryption for persisted signing keys, controller-level **engine hash pinning**, and **IPC bearer tokens** to secure local engine communications. Economically, the protocol utilizes a three-tier system— **KALKI, REDS, and mBITS**—governed by deterministic **KALKI+** issuance and **Quadratic Stake Weighting** to ensure that conviction always outweighs pure capital.

2. Motivation: Three Unsolved Problems

The distributed ledger landscape has historically been defined by a fundamental trade-off between security achieved through computational waste and security derived from capital concentration. Red Shield identifies and addresses three structural challenges inherent in these legacy ecosystems:

2.1 Problem 1 — Quantum Vulnerability Current blockchains, including Bitcoin and Ethereum, rely exclusively on the Elliptic Curve Digital Signature Algorithm (ECDSA) for transaction integrity. ECDSA is mathematically vulnerable to Shor's algorithm, which a sufficiently powerful quantum computer can execute in polynomial time to derive private keys from public keys. Red Shield solves this by utilizing **ML-DSA-87** (NIST FIPS 204) as its native, consensus-level signature scheme from block zero. This lattice-based approach, rooted in the Module Learning With Errors (MLWE) problem, provides a defense-in-depth against both classical and quantum-computational adversaries.

2.2 Problem 2 — Plutocratic Consensus Legacy Proof-of-Stake (PoS) systems grant validator authority as a direct function of asset volume, creating a self-reinforcing feedback loop where wealth begets authority and authority begets more wealth. Red Shield introduces the **Progressive Trust Score (PTS)** to decouple influence from money. Authority is mathematically bound to identity through merit where scores are updated atomically during block processing based on honest participation. Combined with **Quadratic Stake Weighting**, the protocol mathematically ensures that long-term conviction always outweighs raw capital.

2.3 Problem 3 — Partition Fragility The distributed ledger ecosystem has historically struggled with "Partition Fragility," a systemic vulnerability inherent in protocols that rely on "longest chain" or "heaviest chain" rules for fork resolution. In legacy networks - network partitions are treated as zero-sum events that produce "losers". When a fork resolves in these systems, the transactions on the abandoned branch are often discarded—a phenomenon known as **"reorg drops"**—leading to the silent reversal of confirmed user transactions and a fundamental loss of ledger integrity.

Red Shield solves this through the **Never Halt protocol**, a self-healing directive engineered to ensure the network is entirely zero-loss. Unlike legacy models, Red Shield utilizes **Cumulative Validator Trust**—anchored in immutable ML-DSA-signed state roots accumulated over time—rather than simple chain length to determine the legitimate ledger path.

When a fork is detected, the protocol executes a specialized **6-step resolution algorithm**:

1. **Find Common Ancestor:** Identifying the last matching block hash between the divergent chains.
2. **Collect Orphaned Transactions:** Harvesting every user REDS transaction from the blocks being abandoned.
3. **Compare Chain Trust:** Determining the winner based on cumulative Progressive Trust Scores (PTS) rather than height.
4. **Rollback:** Reverting the local chain to the common ancestor height.
5. **Sync From Peer:** Downloading and processing the winning chain branch.
6. **Resubmit Orphaned Transactions:** Re-adding harvested transactions to the mempool for natural inclusion in upcoming blocks.

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3. Protocol Architecture — The SIMBA PRIME

The Red Shield consensus and governance layer is governed by the **SIMBA PRIME** model, an architectural framework designed to solve the structural vulnerabilities of legacy "one-CPU-one-vote" (PoW) or "one-coin-one-vote" (PoS) protocols. SIMBA ensures that the network is controlled by its most reliable and honest participants rather than those with the most capital or hardware.

3.1 Self-Sovereign Node Identity (SSNI)

Unlike permissioned networks, Red Shield validators generate their own cryptographic identities without a central authority. This is achieved through the **Trinity Identity Model**, which mathematically separates a node's identity into three roles to minimize the attack surface:

- **Mother Wallet:** The cold identity derived from a machine-specific master secret; it never signs at runtime.
- **Hot Key (SSNI):** An ephemeral hybrid key pair (Ed25519 + ML-DSA-87) generated on boot that resides in RAM to sign all live attestations.
- **Reward Address:** A decoupled payment destination for block rewards, isolating liquid assets from the operational node environment.

3.2 Identity-to-Authority Binding

In Red Shield, authority is bound to identity through the **Progressive Trust Score (PTS)**, ensuring influence is a result of **merit**, not capital deposits. This score is updated **atomically during block processing** based on verifiable contributions to network health.

Trust Score Gain Schedule

Action	Base Gain	Strategic Utility
Valid PoN Attestation	+1.0	Continuous state verification
State Root Match	+2.0	Confirms local state integrity
Block Production	+5.0	Primary contribution to ledger growth
Emergency Takeover	+20.0	Critical network preservation action

3.3 Merit-Based Validator Promotion

Authority scales through a four-tier status system. Promotion is **self-sovereign**—it occurs automatically when a node crosses a PTS threshold and meets production-hardened uptime requirements.

Status Tier	PTS Threshold	Capabilities	Weight
Active	≥ 50.1	Block production rights	1.0x
Guardian	≥ 70.1	Emergency takeover rights	1.5x

Note: The production-ready **SIMBA_PRIME_3.1)** implementation enforces a mandatory **100-block merit gate** (~50 minutes) for promotion to ensure historical stability.

3.4 Behavioural Attestation (PoN)

While Proof of Capacity (PoC) determines who *proposes* a block, the **Proof of Network (PoN)** layer determines whether that block is *accepted*. Validators create a **ValidatorAttestation**—a dual-signed (ML-DSA + Ed25519) endorsement of the current **State Root**. Finality is achieved only when the network reaches behavioral consensus on the ledger's state.

3.5 Adaptive Consensus Scaling

To maintain security at a small scale while enabling high throughput at a large scale, the required attestation threshold for finality is **adaptive**. It scales logarithmically according to the following production formula:

Formula 1 — Adaptive Threshold

$$Threshold = \frac{K}{log_2(N)}$$

Where K=4.45 and N is the total validator count.

This threshold is capped at **67%** (to ensure Byzantine fault tolerance in small sets) and floored at **51%** (ensuring a strict majority is always required).

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4. Cryptographic Foundation

Red Shield adopts a **"Quantum-Native"** design philosophy, where the cryptographic stack is built on post-quantum primitives from the first line of code rather than being retrofitted onto classical foundations. This ensures the network is secure against Shor’s algorithm and other quantum-computational threats from block zero.

4.1 Universal Hashing: SHAKE-256

All protocol-level hashing utilizes **SHAKE-256** (NIST FIPS 202), an extendable-output function (XOF) from the SHA-3/Keccak family. By replacing legacy SHA-256, Red Shield provides **256-bit Grover-resistant security** across all operations, including transaction identifiers, Merkle tree construction, and Proof of Capacity scoop generation. Using a single, consistent cryptographic family reduces the cryptanalytic attack surface, as an adversary must break Keccak-family primitives rather than multiple disparate algorithms.

4.2 Native Signing: ML-DSA-87

The protocol's primary consensus anchor is **ML-DSA-87** (Module-Lattice Digital Signature Algorithm), standardized as NIST FIPS 204. Rooted in the **Module Learning With Errors (MLWE)** problem, it provides NIST Security Level 5 protection, making it mathematically equivalent to AES-256 in terms of post-quantum strength.

4.3 Hybrid Dual-Signature Model

To provide defense-in-depth and high-performance verification, Red Shield employs a hybrid model where every block, attestation, and consensus-critical message carries **two signatures**:

1. **ML-DSA-87:** The quantum-resistant anchor.
2. **Ed25519:** A high-speed classical signature for fast verification on non-consensus performance paths.

Domain separation is strictly enforced to prevent cross-protocol replay attacks.

Formula 2 — Dual-Signature Validation

$$ValidBlock = ML\_DSA.Verify(Hash, Sig_{pq}) \land Ed25519.Verify(Hash, Sig_{cl})$$

A block is only considered valid if **both signatures** successfully verify over the same SHAKE-256 hash, ensuring the ledger remains immutable even if one signature scheme were eventually compromised.

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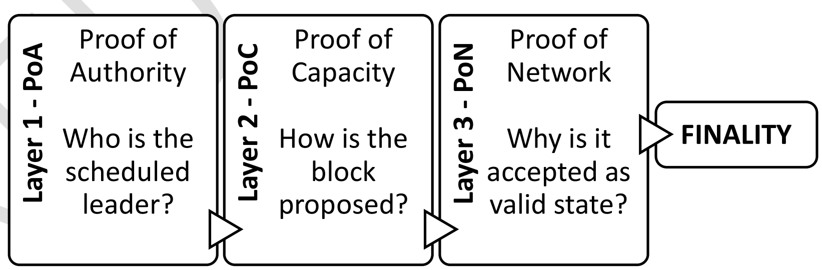
5. Tri-Layer Consensus Architecture

Red Shield prevents single-axis dominance—where capital, hardware, or identity alone can control the ledger—by utilizing a **Tri-Layer Consensus** model. This architecture separates **scheduling**, **proposal**, and **acceptance**, ensuring that for any block to reach finality, it must satisfy three distinct proofs.

Figure 1 — Tri-Layer Consensus Flow



5.1 Layer 1: Proof of Authority (PoA) — Leader Scheduling

Unlike legacy chains where any node can broadcast a block at any time, Red Shield utilizes a **deterministic leader selection** mechanism. For every block height (N), the active validator set from the previous block (N-1) is queried and sorted by address. The leader is selected using the formula: $Leader = Height(\mathrm{mod}\, ActiveValidatorCount)$ This ensures the network knows exactly which node is authorized to initiate the current block round, preventing "spam" proposals and ensuring state stability.

5.2 Layer 2: Proof of Capacity (PoC) — Storage-Based Proposal

Once scheduled, the PoA leader broadcasts a **mining challenge** derived from the previous block hash via SHAKE-256. Validators scan their pre-computed plot files (consisting of iterative SHAKE-256 "scoops") to find the lowest "deadline" value for the challenge.

- **The Mining Power:** A validator's chance of winning block production is proportional to their storage volume and their **conviction multiplier**.
- **Effective Deadline:** The raw deadline is adjusted by the square root of the validator's conviction:

Formula 3 — Effective Deadline

$$EffectiveDeadline = \frac{RawDeadline}{\sqrt{1+LockupYear^{2}}}$$

5.3 Layer 3: Proof of Network (PoN) — Behavioral Attestation

The PoC proposal only provides the *content* of the block; the **Proof of Network (PoN)** layer determines block **finality** through state convergence. Every validator creates a **ValidatorAttestation**—a dual-signed (ML-DSA + Ed25519) endorsement of the current **State Root** (the SHAKE-256 Merkle root of all account balances).

For a block to achieve finality, it must accumulate attestations from a sufficient fraction of the validator set. This required threshold is **adaptive**, scaling logarithmically to maintain security as the network grows.

Formula 4 — Adaptive PoN Threshold

$$Threshold = min(67\%, max(51\%, \frac{K}{log_2(N)}))$$

Where K=4.45 and N is the total validator count. The threshold remains 50% during the initial bootstrap phase (height < 100 or validators < 3).

- ***K* (Scaling Constant):** Fixed at **4.45**.
- ***N* (Validator Count):** The total number of active validators in the set.
- **Cap (67%):** Limits the maximum required consensus to ensure small validator sets can still achieve finality.
- **Floor (51%):** Ensures a strict majority is always required, regardless of how large the network becomes

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6. The KALKI Monetary System

Red Shield utilizes a deterministic and highly precise monetary architecture designed to serve as both a high-frequency transactional currency and a large-scale store of value. The system is built on an **integer-only ledger**, performing all arithmetic in **unsigned 64-bit integers** to eliminate the risk of IEEE 754 floating-point drift and ensure identical state results across all CPU architectures.

6.1 Three-Tier Denomination Architecture

To accommodate different scales of economic activity, the protocol implements a three-tier structure. This allows the network to settle micro-transactions with high precision while remaining readable for major value transfers.

Denomination	Symbol	Conversion	Purpose
mBITS	mBITS	Base Unit	The precision layer. All on-chain balances and settlement operations occur in mBITS.
REDS	REDS	1 = 1,000,000 mBITS	The primary everyday currency used by participants for sending, receiving, and staking.
KALKI	KALKI	1 = 100,000 REDS	The large denomination tier, designed to express major value without long strings of zeros.

The complete mathematical relationship is:
1 KALKI = 100,000 REDS = 100,000,000,000 mBITS.

6.2 KALKI+ — Protocol-Native Issuance

Red Shield rejects centralized minting and foundation-led issuance. New currency enters the ecosystem exclusively through **block rewards**, identified on-chain by the symbolic sender **KALKI+**.

- **Heartbeat Issuance:** Every ~30 seconds, the protocol creates new REDS as a reward for the active block producer and the staking pool.
- **Predictable Schedule:** Issuance is fixed in the source code with no "halvings" or discretionary adjustments. Annual issuance is approximately **~1,051.2 REDS**, ensuring a transparent and deterministically calculable supply at any block height.

6.3 Genesis Allocation and Issuance Logic

At block zero, the protocol initializes its economic reserve through the **Mother Wallet**.

- **The Mother Wallet:** This is the first identity created during genesis initialization. It is credited with the protocol's **FullSupply** (one trillion REDS expressed in mBITS).
- **Economic Reserve:** This total supply serves as the pool for decentralized validator onboarding, network incentives, and bootstrapping the initial validator topology.
- **Distribution Split:** Each per-block reward of **1,000 mBITS** is split to reward both contribution and commitment:
    - **80% (800 mBITS):** Block Producer Reward, sent to the producer's Reward Address.

  - **20% (200 mBITS):** Staking Pool, distributed proportionally to all stakers via **Conviction-Weighted Stake**.
- **Zero-Dust Policy:** A rounding correction algorithm ensures that 100% of the staking pool is distributed in every round, eliminating fractional "dust" losses.

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7. Anti-Whale Economics

Red Shield identifies plutocratic consensus as a structural flaw in legacy Proof-of-Stake systems, where wealth-authority feedback loops inevitably lead to centralized control by the largest capital holders. To ensure the network remains merit-driven and decentralized, the protocol implements **Anti-Whale Economics** through two layers of mathematical dampening designed to ensure that **conviction always outweighs capital**.

Layer 1: Quadratic Stake Weighting

The primary defense against capital dominance is the application of the **Quadratic Stake** formula to all staked assets. Before calculating rewards or voting influence, the protocol computes the deterministic integer square root (**ISQRT**) of the participant's raw stake in mBITS. This creates a system of diminishing returns on influence: a staker with 100x the capital of another participant receives only 10x the pool weight. At extreme scales, a participant with 10,000x the raw capital only achieves a 100x influence ratio, significantly flattening the power curve compared to traditional "one-coin-one-vote" ledgers.

Layer 2: The Conviction Multiplier

While the influence of raw capital is dampened quadratically, long-term commitment is rewarded exponentially. Participants who lock their REDS for a specified duration earn a **Conviction Multiplier** applied to their weighted stake

Formula 5 — Conviction Multiplier

$$ConvictionMultiplier = 1 + (LockupYears)2$$

This multiplier is capped at **10 years**, granting a maximum boost of **101x** to trust gains and staking weight. To maintain protocol-wide determinism, these calculations are performed using architecture-independent integer arithmetic, eliminating the risk of state drift across different validator hardware.

The Meritocratic Result

The synergy of these two layers allows dedicated, small-scale participants to mathematically out-compete liquid "whales". For instance, a small staker with 100,000 mBITS locked for 4 years achieves a total weighted stake of **5,372**, whereas a whale with 100x that capital (10,000,000 mBITS) but no lockup achieves only **3,162**. This structure ensures that those who secure the network's long-term future—rather than those hold the greatest authority and receive the highest rewards. Furthermore, this conviction weight powers the **Time-Shield Protocol**, granting committed validators a structural advantage in Proof of Capacity mining deadlines.

8. Release (SIMBA_PRIME_3.1)

The **Public Release Architecture (SIMBA_PRIME_3.1)** represents the production-hardened distribution model of Red Shield, specifically engineered to lower the barrier for entry while maintaining maximum operational security. It is delivered as a signed **Windows MSI installer** built via a specialized WiX pipeline. This deployment has been manually reviewed and whitelisted by the **Microsoft Security Intelligence Team** to ensure a seamless onboarding experience free from false-positive antivirus detections related to its advanced DPAPI hardware key binding.

A defining feature of this architecture is the **Controller/Engine Split**. The user interacts with a Flutter-based GUI Controller that manages a headless Go backend engine. To secure the local communication channel, the controller injects an ephemeral **IPC Bearer Token** (RED_SHIELD_IPC_TOKEN) into the engine upon startup. This ensures that only the authorized controller can access sensitive node control routes and wallet functions, effectively protecting the node from unauthorized external process interference.

Security is further reinforced through **Integrity Pinning**. The controller embeds the exact **SHA-256 hash** of the production RedShieldEngine.exe and verifies it before every launch. If the engine binary has been tampered with or replaced, the system refuses to boot. Finally, the architecture enforces **Hardened Public Onboarding** safety. To prevent network fragmentation, the onboarding wizard mandates the registration of a **Reward Wallet** and requires a connection to at least one **non-local bootstrap peer**. This ensures every new public node joins the legitimate redshield-mainnet as a stable follower rather than an accidental fork.

9. Production Security Hardening

Red Shield introduces "Machine-Bound" security for public nodes:

- **Windows DPAPI:** Signing keys are encrypted using Windows Data Protection API, binding them to the specific machine and OS user.
- **IPC Middleware:** Local API routes are protected by the RED_SHIELD_IPC_TOKEN, preventing unauthorized external control of the engine.
- **Rate Limits:** The API enforces strict limits (e.g., 100 requests/hour per IP) to prevent DDoS and abuse.

Parameter	Value	Notes
Block Time Target	30 seconds (15 Secs in BetaNet-v1.1.0 – 2 Nodes Setup)	PoC deadline window
Coinbase Reward	1,000 mBITS per block	Issued as KALKI+
Block Producer Share	80% (800 mBITS)	Sent to Payout Address
Staking Pool Share	20% (200 mBITS)	Distributed by conviction weight
Blocks Per Year	~1,051,200	At 30s block time
Annual Issuance	~1,051.2 REDS	~0.01 KALKI per year

Daily Send Limit (Wallet)	10,000 REDS	Client-side, 24h rolling reset
Attestation Max Age	24 hours	Older attestations rejected
Min Consecutive Attestations for Promotion	100	Gate for self-sovereign promotion
Min Uptime for Promotion	~55 minutes	Gate for self-sovereign promotion
Conviction Cap	10 years	Maximum lockup = 101× multiplier
Chain ID	`redshield-mainnet`	Embedded in all PoN snapshots
PoN Protocol Version	1	`/redshield/pon/1.0.0`
Signature Scheme	ML-DSA-87 + Ed25519	Hybrid dual-signature
Hash Function	SHAKE-256	NIST FIPS 202
DB Backend	BadgerDB	Embedded key-value store
P2P Layer	libp2p	DHT + gossip
Active Threshold	≥ 50.1 PTS	Block production
Guardian Threshold	≥ 70.1 PTS	Emergency takeover, 1.5× voting

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10. Roadmap

Phase	Milestone	Status
Phase 1 — Foundation	Protocol design, SIMBA consensus, ML-DSA integration	✅ Complete
Phase 2 — BetaNet	Two-node deployment, wallet release, performance audit	✅ Complete (v1.1.0)
Phase 3 — Expansion	Scale to 10+ validators, stress-test adaptive PoN threshold	In Progress
Phase 4 — Public MainNet	Open validator registration, community node deployment. Production launch, full REDS economy activation	Planned
Phase 5 — Ecosystem	Interoperability	Planned

11. Conclusion

Red Shield is not a conceptual promise but a **running network**. By combining hybrid post-quantum cryptography with merit-based consensus, it provides a stable, energy-efficient, and mathematically fair alternative to legacy blockchains. The deployment of **SIMBA_PRIME_3.1** marks the transition from BetaNet to a production-hardened public ecosystem where authority is truly earned.

Red Shield introduces a new class of distributed ledger — quantum-native, behaviour-driven, self-healing, and economically fair. The SIMBA protocol and the REDS currency system compose into a network where:

- *Authority is earned* through honest participation, not purchased through capital.
- *The cryptographic foundation* resists both classical and quantum adversaries from block zero.
- *Network partitions heal* without user loss — no REDS are ever silently reversed.
- *New validators join autonomously*, verify independently, and earn trust incrementally.
- *Currency issuance is transparent*, fixed, and controlled by mathematics — not by any institution.
- *Economic fairness is enforced mathematically* through quadratic stake weighting and conviction-based rewards.

The BetaNet v1.1.0 deployment confirms that all protocol properties function as designed. Block production is stable. Both validators achieved GUARDIAN status through merit alone. ML-DSA post-quantum signatures protect every block. REDS are issuing on schedule via KALKI+. The Never Halt fork resolution is deployed and operational.

Appendix A — Glossary

Term	Definition
Attestation	A cryptographically signed endorsement of the current network state, broadcast by a validator during each block round.
BadgerDB	An embedded, persistent key-value database (v4) used by Red Shield nodes for chain storage and state management.
Conviction Multiplier	A mathematical factor (`$1 + \text{years}^2$`) that amplifies a validator's trust gain and staking weight based on their lockup duration.
ECDSA	Elliptic Curve Digital Signature Algorithm — the classical signing scheme used by Bitcoin, Ethereum, and most existing blockchains. Vulnerable to Shor's algorithm.
Ed25519	A high-performance classical signature scheme (Edwards-curve Digital Signature Algorithm). Used

	by Red Shield as the secondary signature in its hybrid model.
Effective Deadline	A validator's PoC mining deadline adjusted by the Time-Shield Protocol's conviction factor. Lower values win block production.
FIPS 202	NIST standard defining SHAKE-256 and other SHA-3/Keccak hash functions.
FIPS 204	NIST standard defining ML-DSA (Module-Lattice Digital Signature Algorithm). Published 2024.
Genesis Node	The founding node of the Red Shield network. Creates the genesis block and serves as the initial bootstrap peer.
Grover's Algorithm	A quantum algorithm that provides a quadratic speedup for unstructured search problems, reducing the effective security of hash functions by half.
Guardian	The highest validator status tier (PTS ≥ 70.1). Guardians have emergency takeover rights and 1.5× voting weight.
Hot Key (SSNI)	Self-Sovereign Node Identity — the ephemeral signing key generated on first boot. Signs blocks and attestations. Holds no funds.
ISQRT	Integer Square Root — a deterministic, architecture-independent square root function using the Digit-by-digit method. Used for quadratic stake weighting.
KALKI	The large denomination of the REDS currency. 1 KALKI = 100,000 REDS.
KALKI+	The protocol's symbolic identifier for block reward issuance events. Not a wallet address.
libp2p	A modular peer-to-peer networking stack used by Red Shield for node communication, peer discovery, and gossip protocols.
mBITS	The base precision unit of the REDS currency. 1,000,000 mBITS = 1 REDS. All on-chain arithmetic operates in mBITS.
Merkle Root	A single hash that cryptographically commits to the entire set of account balances (state). Computed using SHAKE-256.
ML-DSA	Module-Lattice Digital Signature Algorithm — the NIST-standardised post-quantum signature scheme (FIPS 204).
ML-DSA-87	The highest security parameter set of ML-DSA, providing NIST Security Level 5 (equivalent to AES-256 post-quantum security).

MLWE	Module Learning With Errors — the mathematical hardness assumption underlying ML-DSA's security. No known polynomial-time quantum attack.
Mother Wallet	The cold identity of a validator node. Derived from a machine-specific master secret. Never signs at runtime.
Never Halt	Red Shield's fork resolution directive guaranteeing that the chain never stops and no user transaction is silently lost during partition recovery.
PoC	Proof of Capacity — a consensus mechanism where block leader selection is based on pre-computed plot files stored on disk, replacing energy-intensive computation with storage I/O.
PoN	Proof of Network — Red Shield's attestation-based consensus layer that determines block acceptance through multi-validator endorsement.
PQC_SECURE	A metadata tag embedded in every Red Shield block, asserting that all signatures use the ML-DSA scheme.
Progressive Trust Score (PTS)	A numerical score tracking a validator's historical honesty and reliability. Updated atomically during block processing.
Quadratic Stake	The square root of a validator's raw stake in mBITS. Used to dampen the influence of large capital holders. `QuadraticStake = √(RawStake)`.
REDS	The primary everyday denomination of Red Shield's native currency. 1 REDS = 1,000,000 mBITS.
Reward Address	The third component of the Trinity Identity. A wallet address that receives block rewards, decoupled from the signing identity.
Roaming Node	A follower validator that joins the network via peer discovery, syncs state, registers on-chain, and begins producing blocks.
SHAKE-256	An extendable-output function (XOF) from the SHA-3/Keccak family (NIST FIPS 202). Red Shield's universal hash function.
Shor's Algorithm	A quantum algorithm that solves the discrete logarithm and integer factorisation problems in polynomial time, breaking ECDSA and RSA.
SIMBA	Self-Sovereign Identity, Merit-Based Authority, Behavioural Attestation — the five-pillar consensus and governance model of Red Shield.
State Root	A SHAKE-256 Merkle hash of all account balances at a given block height. Used for trustless synchronisation and attestation verification.
Time-Shield Protocol	The mechanism that adjusts PoC mining deadlines based on √(conviction), allowing committed

	validators to compete against larger storage operators.
Trinity Model	Red Shield's three-key identity architecture: Mother Wallet (cold identity), Hot Key (signing), Reward Address (payment).
Validator	A network participant that produces blocks, signs attestations, and earns REDS through the SIMBA consensus mechanism.

Appendix B — References

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GitHub
https://github.com/redshield-pqc

Releases
1) https://github.com/redshield-pqc/RED-SHIELD---SIMBA-Prime/releases/tag/Quantum_Native_Wallet
2) https://github.com/redshield-pqc/RED-SHIELD---SIMBA-Prime/releases/tag/Post_Quantum_Cryptography
3) https://github.com/redshield-pqc/RED-SHIELD---SIMBA-Prime/releases/tag/SIMBA_PRIME_3.1

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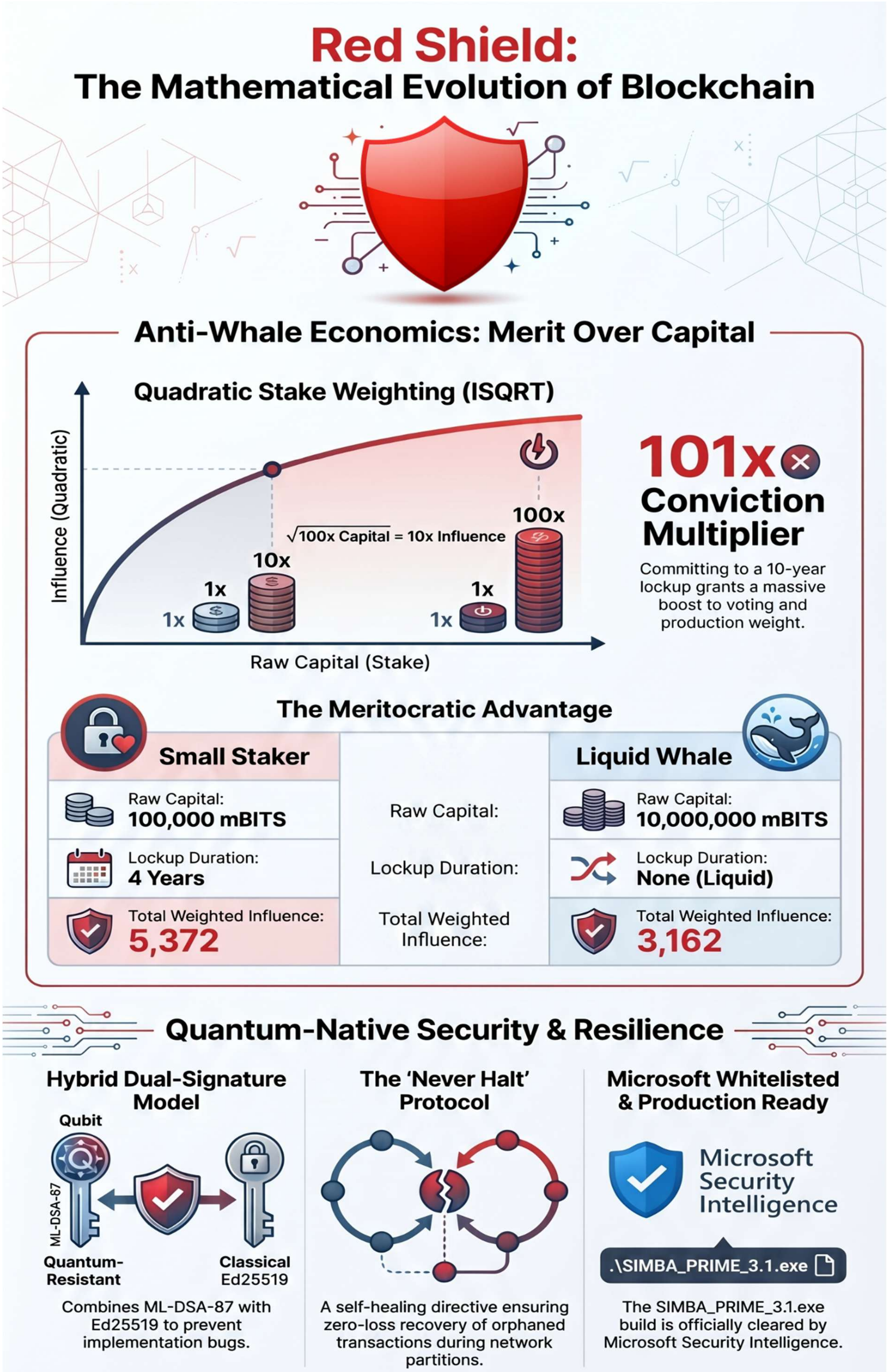

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